



Solace – A Privacy-First End-to-End Encrypted Chat Platform with Team Workspaces

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Abstract

- Solace is a web chat platform in which every message, file, voice note and call signal is encrypted in the sender's browser, so the server stores and routes ciphertext only.
- Any group chat can become a Workspace with tasks, calendar, notes, decisions, polls and files, built on Next.js, Fastify and a single MongoDB datastore.

Problem Statement & Objectives

- Problem:** Mainstream chat and team tools store content the provider can read, and team work is scattered across separate apps.
- Objective 1:** Real-time 1:1 and group messaging in which the server never sees plaintext or private keys.
- Objective 2:** Safe membership changes: a removed member cannot read new messages, with no key rotation.
- Objective 3:** Encrypted Workspaces (tasks, events, notes, polls, files) and peer-to-peer voice/video calls.
- Scope:** Web app for individuals and small teams; roles Owner / Admin / Member; 1:1 calls only; no native mobile app.

Existing vs Proposed System

Feature	Existing System	Proposed System
Message privacy	Provider can read stored content (e.g. Slack)	Encrypted in browser; server holds ciphertext
Team tools	Separate apps for chat, tasks, calendar and files	Any group chat becomes a Workspace in the same app
Workspace data	Task, note and poll text readable by the service	Titles, notes, poll options encrypted client-side
Calls	Call signaling readable by the server	P2P WebRTC, end-to-end encrypted signaling
Limitation / Gap	Private messengers lack team tools; team tools lack E2EE	One app with both, on one MongoDB datastore

Feasibility Study

- Technical:** Open-source TypeScript stack; WebCrypto and WebRTC are built into modern browsers; the full stack runs on one laptop.
- Economic:** No licence cost; free tiers of MongoDB Atlas (M0) and Vercel are enough for development and demos.
- Operational:** Familiar sign-up and chat flow; encryption keys are generated automatically on each device; one datastore to maintain.
- Schedule:** Built in seven incremental phases plus Workspace features within Trimester 4 (2026–27).

Results & Testing

- 60 automated tests: 22 crypto + 38 server
- 37 / 37 crypto, null-parity and media-URL tests pass
- 23 call-signaling tests run on a MongoDB replica set
- CI runs type-check, tests and web build

Conclusion

- Delivered E2EE chat, media, voice notes, calls and encrypted Workspaces
- Server stores only ciphertext, public keys and routing metadata
- Simplified to one datastore: MongoDB replaced PostgreSQL, Redis and MinIO

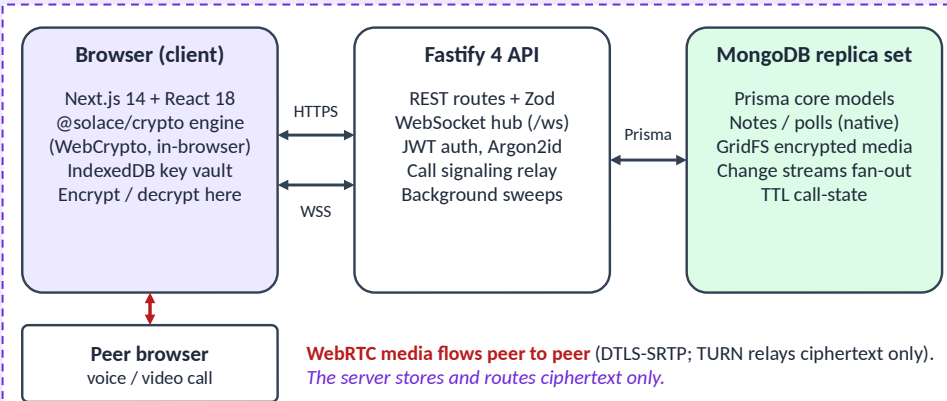
Future Scope

- Forward-secure ratchet on top of the stateless envelope
- Group voice and video calls
- Key-fingerprint (safety number) verification
- PWA with push notifications

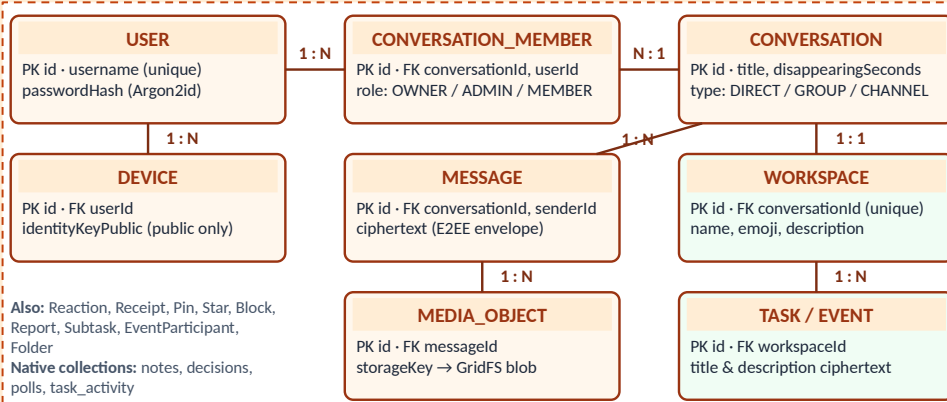
Tools & Technologies

- Front-End:** Next.js 14, React 18, TypeScript, Tailwind CSS
- Back-End:** Node.js + Fastify 4, REST APIs + WebSocket (/ws), Zod
- Database:** MongoDB replica set – Prisma 5, GridFS, change streams
- Security:** WebCrypto (ECDH P-256, HKDF-SHA256, AES-256-GCM), Argon2id, JWT
- Tools:** Git & GitHub, GitHub Actions CI, VS Code, MongoDB Compass, Vercel

System Architecture



ER Diagram & Database Design



Implementation Screenshots

